

Research on Bid Evaluation Data Storage System Based on Blockchain

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Abstract—The traditional electronic bidding platform has some problems, such as easy to tamper with bid evaluation data and opaque bid evaluation process, which leads to the lack of public confidence in the electronic bidding platform. In order to improve the credibility of the electronic bidding platform among the public, this paper uses the alliance blockchain (Hyperledger Fabric) and the Interplanetary file system (IPFS) to build a bid evaluation data storage system. The Raft consensus mechanism ensures consistency and reliability between nodes, ensuring the correctness and trustworthiness of transactions throughout the network. Use smart contracts to monitor and certify bid evaluation data; Develop front-end pages to get through the whole process. The system uses blockchain technology as the underlying application to make the bid evaluation data immutable and traceable, which greatly improves the fairness and transparency of the electronic bidding platform, strengthens the supervision of the government and the public on the electronic bidding platform, and promotes the creation of a fair, just and open bidding environment.

Index Terms—blockchain, IPFS, electronic bidding, evaluation of bid, data storage

I. INTRODUCTION

With the rapid development of information technology, electronic bidding has been able to shine in the Internet. Compared with the traditional way of offline bidding, electronic bidding has great advantages. Through the online bidding and bidding platform, electronic bidding, bidding, evaluation and other work can significantly reduce the time cycle of bidding and inviting bidding, improve the efficiency of bidding and inviting bidding, and reduce human intervention and information asymmetry and other problems, ensure the equal participation of bidders, improve the fairness and transparency of bidding results. At present, China's bidding authorities have issued a lot of policies or measures to encourage the use of electronic bidding, and electronic bidding has grad-

ually become the first choice for most procurement entities to conduct business operations. Bid evaluation is a crucial link in the bidding process, and its data fairness is of great significance to the public. However, bidding collusion and system internal operators bending the law in the face of huge economic temptations [1] are easy to cause data tampering and damage the integrity system, which are the security risks faced by electronic bidding. In order to ensure the security of each platform of the electronic bidding system, it is necessary to consider the implementation of technical means to ensure that the data is not tampered with, not omitted and traceable during the construction of the system.

Nowadays, with the rapid development of blockchain technology, its immutable, distributed storage and traceable characteristics are an ideal choice for electronic bidding platforms to carry out decentralized bid evaluation data storage [2]. In the blockchain network, bidding agencies, bidding companies, bidding companies and regulatory authorities are all in the same blockchain network, and each participant manages a data ledger, so as to realize the joint maintenance and utilization of bid evaluation data across departments and regions [3]. Through the decentralized management model, each participant has the ability to manage the complete data record and jointly verify and record transactions, thus effectively avoiding the problems that can be caused by a single central control.

Therefore, this paper proposes a bid evaluation data storage system based on blockchain to solve the problem of reliable storage of bid evaluation data in electronic bidding platform. Based on the user access mechanism of Hyperledger Fabric [4], as well as the features of security, traceability and immutable, combined with the IPFS interplanetary file system, a data storage system is jointly constructed to realize the safe and trusted storage of bid evaluation data of the electronic

bidding platform.

II. TECHNOLOGY INTRODUCTION

A. Blockchain Technology

Blockchain technology is a decentralized distributed ledger technology that originally emerged as the underlying technology for Bitcoin. Distributed ledger technology marks a revolutionary progress in information technology architecture, which enables transaction data to be recorded, shared and synchronized across geographies by multiple nodes in a decentralized network [5]. It is fundamentally different from traditional centralized ledgers in that it does not rely on any single central management entity, but maintains copies of the ledgers through the joint efforts of various nodes in the network. This distributed architecture ensures transparency, immutability and consistency of transactions. Once data is written to the blockchain, it is almost impossible to tamper with or delete because each block contains the hash of the previous block, forming an irreversible chain link [6]. The blockchain data structure is shown in Figure 1. Because data is encrypted and protected by cryptographic methods, blockchain has a high level of security, making it extremely difficult for data to be stolen or modified during transmission and storage.

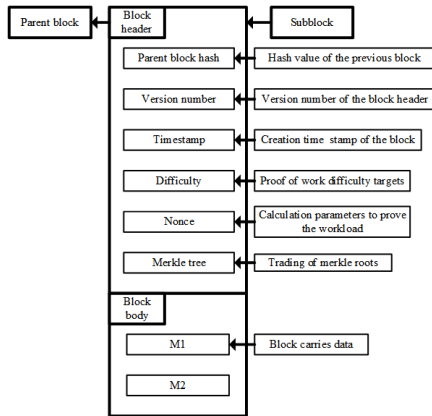


Fig. 1. Blockchain Ledger Structure Diagram

B. IPFS Technology

IPFS uses a distributed storage model [7] where files are split into blocks and stored on various nodes of the network, rather than centrally stored on a single server. Use content addressing to locate and access files, rather than depending on the location or path of the file. Each file and block of files has a unique hash value that is calculated based on the contents of the file, thus ensuring the integrity and uniqueness of the file. IPFS improves performance by replicating and caching blocks of data, allowing users to retrieve data from the nearest node, reducing latency and bandwidth costs. IPFS is an open protocol that allows anyone to use and build on top of it, providing infrastructure support for building decentralized applications. The design of IPFS makes it an ideal infrastructure for decentralized applications, such as

blockchain applications, without relying on a single server or centralized service provider.

C. Smart Contract

Smart contracts are one of the important innovations in blockchain technology, creating a new era of operational mechanisms and transactions. As a program capable of automatic execution and self-verification, smart contracts enable complex transactions and contract terms to be completed automatically without the involvement of third-party intermediaries [8]. Smart contracts define rules and agreements between participants based on code, and at the same time they do not require human operation, as long as the conditions set in the contract are met, the relevant terms will be automatically executed. Smart contracts are unique in that they are decentralized, enabling them to run on any computing node in the world, significantly improving execution efficiency and enhancing system security. The code is transparent and immutable, ensuring the auditability and verification of the contract function, ensuring that the contract strictly follows the logic of the initial design, and the irreversibility of execution guarantees the finality and consistency of the operation.

III. SYSTEM DESIGN

A. System Architecture

The bid evaluation data storage system architecture based on blockchain designed in this paper is shown in Figure 2, which is mainly divided into three layers, namely application layer, intermediate layer and storage layer.

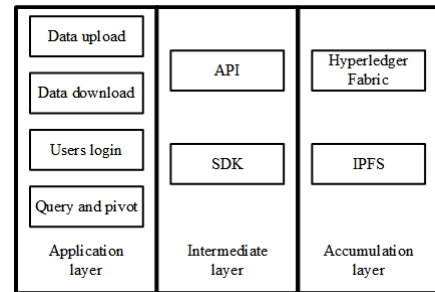


Fig. 2. System Architecture Diagram

- **Application layer:** The application layer provides complete system services to users in the form of interfaces. The application layer can realize the following functions: data upload, data download, data query, and user login.
- **Intermediate layer:** The intermediate layer serves as the transition layer and connects the upper application layer with the lower storage layer. Mainly responsible for the implementation of relevant functional interfaces and data upload and download.
- **Storage layer:** The storage layer contains the deployed IPFS interplanetary file system and the affiliate chain platform. The IPFS interplanetary file system is responsible for two functions: the storage of complete data and the return of data hash summaries; Download the stored data based on the hash summary.

B. Storage Model

After in-depth understanding of the bid evaluation data storage system, a data storage model based on blockchain technology platform is constructed. At present, there are mainly three types of cooperation modes of blockchain technology [9], which are public chain, private chain and alliance chain. Table 1 lists the characteristics of each of the three types of blockchain. Because the system involves multiple organizations, consider a Hyperledger Fabric blockchain network that belongs to a consortium chain.

TABLE I
THREE TYPES OF BLOCKCHAINS HAVE CHARACTERISTICS TABLE

Type	Private Chain	Alliance Chain	Public Chain
Permission	Limited participation	Partial open	Wide open
Participant	Single	Multiple	Unlimited
Safety	Low to medium	Medium to high	High

A bid evaluation data storage system model based on Fabric blockchain network is designed with blockchain technology as the underlying application. Many types of files are involved in the bid evaluation process, including videos and pictures. These files are stored through the IPFS system, and each file block generates a unique summary of the file hash based on its contents. These hash summaries and their upload records will be stored on the blockchain to ensure secure storage of bid evaluation data, data integrity and immutability. This design ensures that the various types of documents involved in the bid evaluation process can be safely and reliably stored in the IPFS system, and their upload records and hash summaries are recorded and verified through blockchain technology, thus preventing data from being tampered with or loss. The bid evaluation data storage system model is shown in Figure 3.

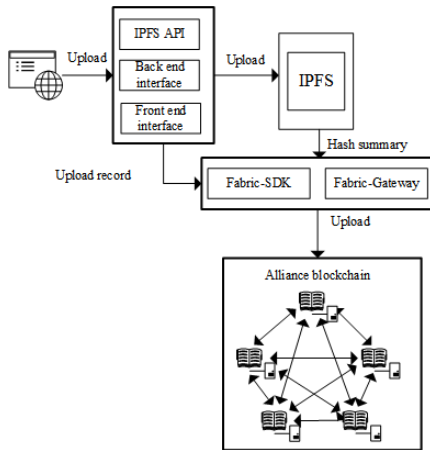


Fig. 3. Storage Model Diagram

The IPFS distributed storage system bears the pressure of data storage, making up for the shortcomings of limited on-chain storage space and low transaction processing efficiency

in the blockchain system, and the deep integration of the two not only retains the advantages of the blockchain itself but also improves the efficiency of data storage.

C. Consensus Mechanism

The adoption of the Raft consensus mechanism in the Fabric blockchain network ensures system stability and fault tolerance [10]. Raft is an algorithm used to implement distributed consensus and is primarily responsible for managing log replication to ensure system consistency. Raft is native to Hyperledger Fabric and directly supported by Hyperledger Fabric, which can better increase system stability and support subsequent iterations of system versions. Meanwhile, Raft allows users to specify which sort node to deploy to which channel. Instead of trusting and relying on an additional organization to manage the sorting and consensus services, node organizations can make a more decentralized system by ensuring that if they also have a sorting node, that node will be part of the sorting service for that channel. However, since Raft consensus is not Byzantine fault-tolerant, once on-chain nodes do evil, it will have a great impact on the blockchain and the entire system.

D. System Client Development

The bid evaluation data storage system takes alliance chain and IPFS as the application base, realizes data interaction with the outside world through smart contracts, and realizes data upload and download through front-end pages. The front-end page is developed using Vue3, Node.js[11], and element-ui. The backend utilizes the SpringBoot framework to interact with the IPFS system through the IPFS external interface, and integrates with the blockchain network using the Java SDK provided by the Hyperledger Fabric network. The front-end function module of the system includes personal center, upload file, download file and data query function. The personal center module is responsible for user account registration, password management and system login; The upload file module allows users to upload bid evaluation information, which is processed by back-end logic and submitted to the blockchain network; The download file module obtains the hash summary by the blockchain, retrieves and downloads the qualified files in the IPFS system, and displays the relevant bid evaluation data; The data query feature allows users to filter and find records of bid evaluation data stored on the blockchain.

E. System Operation Flow

In practical applications, the process of bid evaluation data is as follows (see Figure 4) : The bid evaluation data is collated by the manager and uploaded to the deployed IPFS system node. The system generates a unique hash summary based on the content of the data and returns it to the user. The hash summary is then encrypted with a key and operated on a blockchain via a smart contract. This process ensures that the data is secure and immutable. When downloading the bid evaluation data, the user obtains the hash summary ciphertext from the blockchain through a smart contract and performs

key decryption to get the plaintext of the hash summary. Then, the system uses the hash value to retrieve the corresponding bid evaluation data in the IPFS system, so as to complete the data acquisition and display. All bid evaluation data downloads and uploads are recorded on the consortium blockchain, ensuring the transparency and traceability of the operations. If any error occurs during the operation, the smart contract will provide the corresponding error return information for timely processing and correction. The smart contract designed in this way fully reflects the idea of blockchain decentralization, and provides support and guarantee for important technical tools in practical applications.

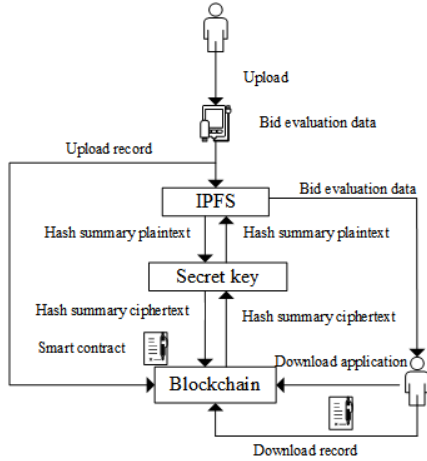


Fig. 4. System Operation Flow Diagram

F. Blockchain System Performance Analysis

After the evaluation data storage system is developed, the credibility test of the system is carried out to ensure that the evaluation data can be successfully stored in the system and can be correctly queried in the system. In the course of testing, gradually increase the number of transactions, and record the number of successful writes and reads. Table 2 shows the results of the test. The transaction is always in a successful state and can be successfully queried, which means that the system has a high credibility and can be applied to the e-bidding platform. Once the bid evaluation data is recorded on the blockchain, it cannot be modified or erased, ensuring the immutability and long-term preservation of the data. By combining the immutability of the blockchain system and the reliability of the test system, the immutability and traceability of the bid evaluation data in the system are guaranteed.

TABLE II
SYSTEM RELIABILITY TEST RESULTS TABLE

Number of transactions	200	400	600	800	1000	1200
Number of successes	200	400	600	800	1000	1200

IV. CONCLUSION

The reliable storage of bid evaluation data is a key part of building a fair and just e-bidding environment. In this paper, a bid evaluation data storage system based on blockchain technology is designed to solve the drawbacks of bid evaluation data storage in combination with the electronic bidding platform and the characteristics of blockchain and IPFS technology, providing a certain reference case for future research on the application of blockchain technology to electronic bidding platform. However, considering from many aspects, there is still a large room for improvement. Subsequently, the blockchain technology can be combined with the entire process of electronic bidding, and the chain operation can be carried out at all stages of bidding to ensure that the bidding process can be traced throughout. The bid evaluation data storage system based on blockchain technology can effectively solve the storage of the bid evaluation data of the electronic bidding platform, and at the same time, the bid evaluation data can be traced to ensure the fairness and transparency of the electronic bidding platform.

REFERENCES

- [1] Long W, Wang X, Liang Y, et al. Characteristics of collusive practices in bidding: Mixed methods study in China[J]. *Journal of Construction Engineering and Management*, 2023, 149(6): 04023030.
- [2] Yadav A S, Singh N, Kushwaha D S. Evolution of Blockchain and consensus mechanisms & its real-world applications[J]. *Multimedia Tools and Applications*, 2023, 82(22): 34363-34408.
- [3] Barger A, Funaro L, Laventman G, et al. Orion: A Centralized Blockchain Database with Multi-Party Data Access Control[C]//2023 IEEE International Conference on Blockchain and Cryptocurrency (ICBC). IEEE, 2023: 1-9.
- [4] Jyothilakshmi K B, Robins V, Mahesh A S. A comparative analysis between hyperledger fabric and ethereum in medical sector: A systematic review[J]. *Sustainable Communication Networks and Application: Proceedings of ICSCN 2021*, 2022: 67-86.
- [5] Soltani R, Zaman M, Joshi R, et al. Distributed ledger technologies and their applications: A review[J]. *Applied Sciences*, 2022, 12(15): 7898.
- [6] Truong N B, Sun K, Lee G M, et al. Gdpr-compliant personal data management: A blockchain-based solution[J]. *IEEE Transactions on Information Forensics and Security*, 2019, 15: 1746-1761.
- [7] Benet J. IPFS-content addressed, versioned, P2P file system (DRAFT 3)[J]. *arXiv preprint arXiv:1407.3561*, 2014: 1-11.
- [8] Wang, S., Ouyang, L., Yuan, Y., et al. (2019) Block-chain-Enabled Smart Contracts: Architecture, Applications, and Future Trends. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 49, 2266-2277.
- [9] Paul P, Aithal P S, Saavedra R, et al. Blockchain technology and its types—a short review[J]. *International Journal of Applied Science and Engineering (IJASE)*, 2021, 9(2): 189-200.
- [10] Fu W, Wei X, Tong S. An improved blockchain consensus algorithm based on raft[J]. *Arabian Journal for Science and Engineering*, 2021, 46(9): 8137-8149.
- [11] Kumar N, Dakshayani M. Secure sharing of health data using hyperledger fabric based on blockchain technology[C]. *2020 International Conference on Mainstreaming Block Chain Implementation (ICOMBI)*. Bengaluru, India: IEEE, 2020: 1-5.